GAD vs Sella vs Hybrid GAD–Newton

Comprehensive TS-finder benchmark on Transition1x test split

2026-05-11. $n = 287$ samples per (method, noise). Best-of-family comparison.

TL;DR

Three families benchmarked at 6 noise levels (10–200 pm) on T1x test: plain GAD (a Hessian-following Euler integrator), Sella (a trust-region Newton step in internal coordinates), and the hybrid GAD–Newton step (GAD walks, then Newton lands when the curvature signature is right). Best of each family across noise:

> **Sella libdef** wins raw conv ≤50 pm (92.7% @ 10 pm). **Plain GAD dt=0.005** wins IRC TOPO at ≥100 pm noise (+11.9 pp over Sella at 150 pm; +21.3 pp at 200 pm). **Hybrid damped Eckart eig-switch tr=0.05** wins *wall-time per converged TS* at every noise level (1.4–3.0× vs Sella; 4.1–4.5× vs plain GAD), and produces *geometrically tighter* converged TSs than either baseline at high noise (p95 RMSD to known TS at 200 pm: hybrid 0.11 Å vs Sella 0.84 Å vs plain GAD 0.46 Å).

Recommendation.

Goal	Use	Why
Fastest TS finder, any noise	`hybrid_damped_eckart_swtrue` tr=0.05	1.4–3× vs Sella
Highest IRC TOPO at ≥100 pm	Plain GAD dt=0.005	Right-basin failures
Tightest geometry at high noise	`hybrid_damped_eckart_swtrue` tr=0.05	7.7× tighter p95
Low-noise raw conv (≤50 pm)	Sella libdef	92.7% @ 10 pm

1 Headline comparison

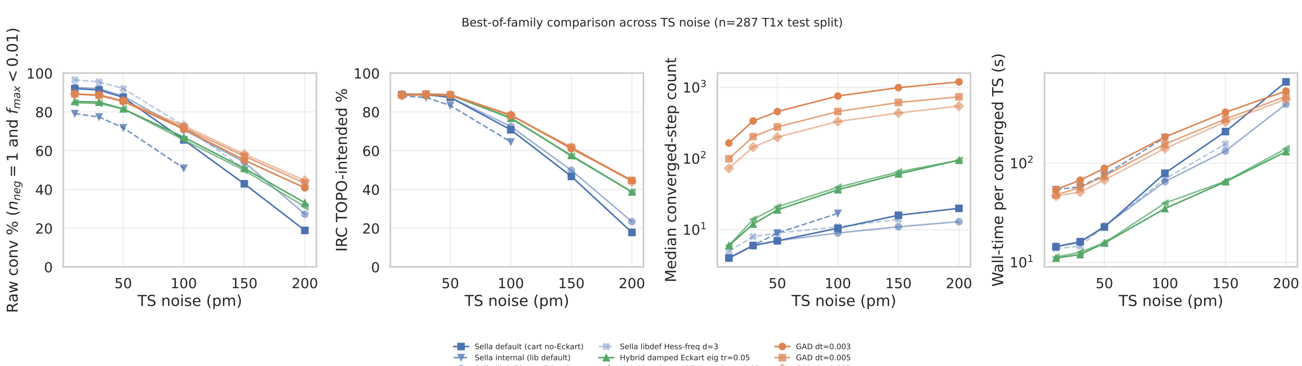

Figure 1: Best-of-family across noise. Four diagnostic axes: raw conv, IRC TOPO, median converged steps, wall-time per converged TS. Each family contributes its strongest configs as separate lines. The Sella libdef Hess-freq d=3 variant (only raw conv available, no IRC) shows up only on panel 1. Sella internal only goes to 100 pm. All others span 10–200 pm. Source: `master_2026_05_11.csv`.

Per-noise tables. All numbers from `analysis_2026_04_29/master_2026_05_11.csv`. Bold = best in row.

Table 1 — Raw conv % (n_{neg}=1 $\wedge$ $f_{\text{max}} < 0.01$, $n = 287$)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.003	89.2	88.5	85.4	71.1	55.1	40.8
GAD dt=0.005	89.2	88.5	85.7	71.8	57.1	43.2
GAD dt=0.007	89.2	88.9	85.7	72.8	58.2	**44.6**
Sella default (cart no-Eckart)	92.0	91.3	87.5	65.5	42.9	18.8
Sella libdef (cart+Eckart)	92.7	92.0	88.2	70.7	54.0	27.2
Sella internal	79.1	77.4	71.8	50.9	—	—
Sella libdef Hess-freq d=3	**96.5**	**95.5**	**92.0**	72.8	50.5	—
Hybrid damped Eckart eig tr=0.05	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped Eckart eig tr=0.05	84.7	84.3	81.5	65.5	49.8	31.0

Table 2 — IRC TOPO-intended % (Sella IRC graph-matches reactant+product, $n = 287$)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.003	88.5	**89.2**	88.9	78.4	61.0	44.6
GAD dt=0.005	88.9	**89.2**	88.9	78.0	61.7	**44.6**
GAD dt=0.007	88.5	88.5	88.5	**78.4**	**61.7**	43.9
Sella default	88.9	88.9	87.5	70.7	46.7	17.8
Sella libdef	**89.2**	**89.2**	87.5	72.5	49.8	23.3
Sella internal	88.2	87.5	83.3	64.5	—	—
Sella libdef Hess-freq d=3	—	—	—	—	—	—
Hybrid damped Eckart eig tr=0.05	**89.2**	88.9	88.9	76.7	57.5	38.7
Hybrid undamped Eckart eig tr=0.05	88.9	88.9	88.5	77.0	57.1	38.7

Table 3 — IRC RMSD-intended % (strict 0.3 Å threshold, $n = 287$)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.005	46.3	46.7	45.6	**40.1**	**31.4**	**21.3**
Sella libdef	45.6	46.3	44.6	36.6	24.7	10.1
Hybrid damped Eckart eig tr=0.05	46.0	46.0	44.9	37.6	28.2	18.1

2 Ranking by wall-time per converged TS

The headline 4-axis figure shows trends; the next two visualizations make rankings concrete.

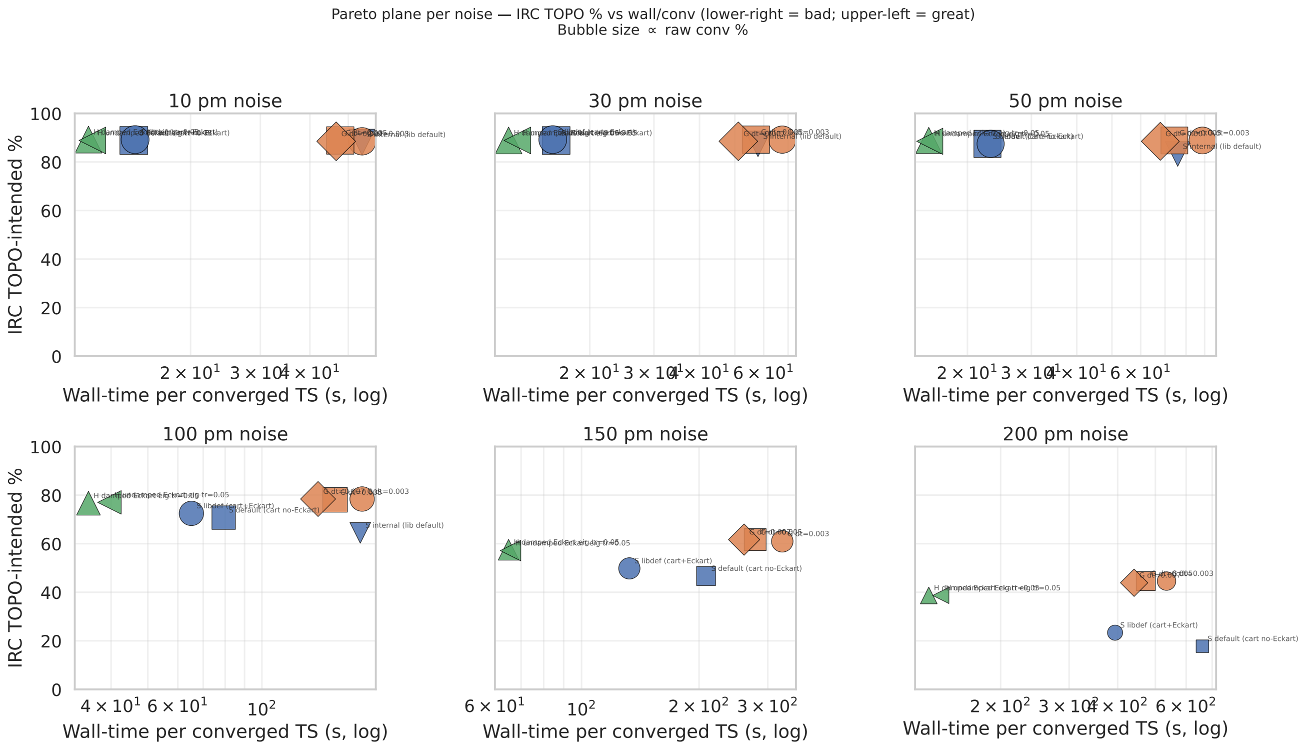

Figure 2: Pareto plane per noise. x = wall-time per converged TS (log); y = IRC TOPO. Bubble size ∝ raw conv. Upper-left = great (fast AND chemistry-correct); lower-right = bad (slow AND chemistry-failing). Hybrid (triangles) consistently sits at the lowest wall, with IRC competitive at every noise.

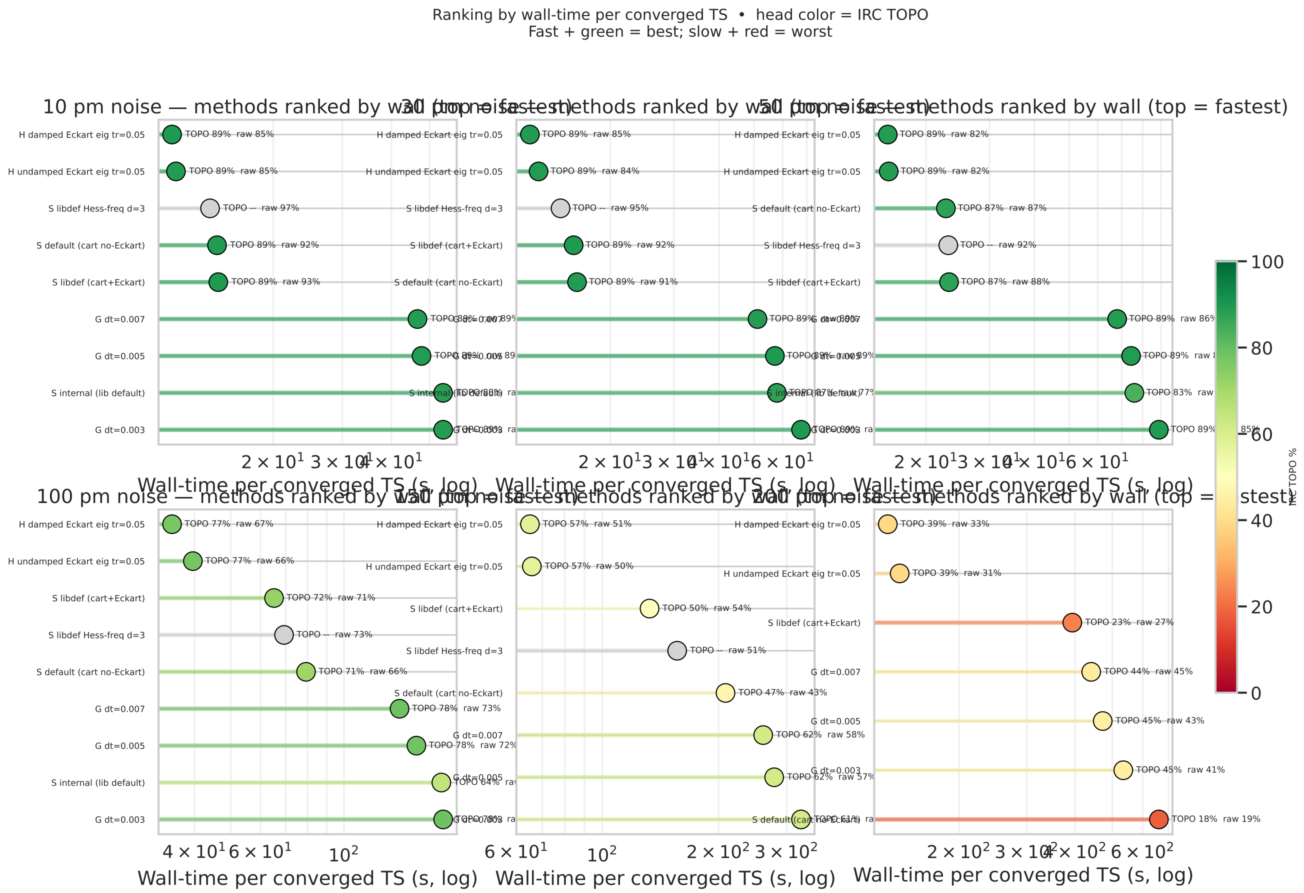

Figure 3: Methods ranked by wall/conv per noise. Each panel sorts methods by wall-time ascending (top = fastest). Head color = IRC TOPO % (red = bad chemistry, green = good chemistry). Inline annotations: TOPO and raw conv. Across all 6 noises, the hybrid configurations occupy the top of the ranking with green heads. Sella libdef Hess-freq d=3 (no IRC) shows as gray heads.

Table 4 — Wall-time per converged TS (s)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.005	47.7	57.1	74.3	156.0	278.5	472.3
Sella libdef	14.5	15.8	23.2	65.2	132.6	393.8
Sella libdef Hess-freq d=3	**13.8**	**14.6**	23.1	69.3	156.3	—
Hybrid damped Eckart eig tr=0.05	**11.0**	**12.0**	**15.7**	**34.9**	**65.0**	**130.7**

Table 5 — Median converged-step count

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.005	100	204	278	458	614	738
Sella libdef	**4**	**6**	**7**	**9**	**11**	**13**
Sella libdef Hess-freq d=3	5	8	9	11	14	—
Hybrid damped Eckart eig tr=0.05	6	12	19	37	61	95

Reading the ranking. Sella has the fewest steps, but each Sella step is expensive (multiple Hessian-update operations). The hybrid takes 4–7× more steps than Sella but each step is cheaper, so net wall/conv is lower. Plain GAD takes 70–700× more steps than Sella and pays for all of them — slowest of the three families.

3 Why GAD and hybrid beat Sella on chemistry (mechanism)

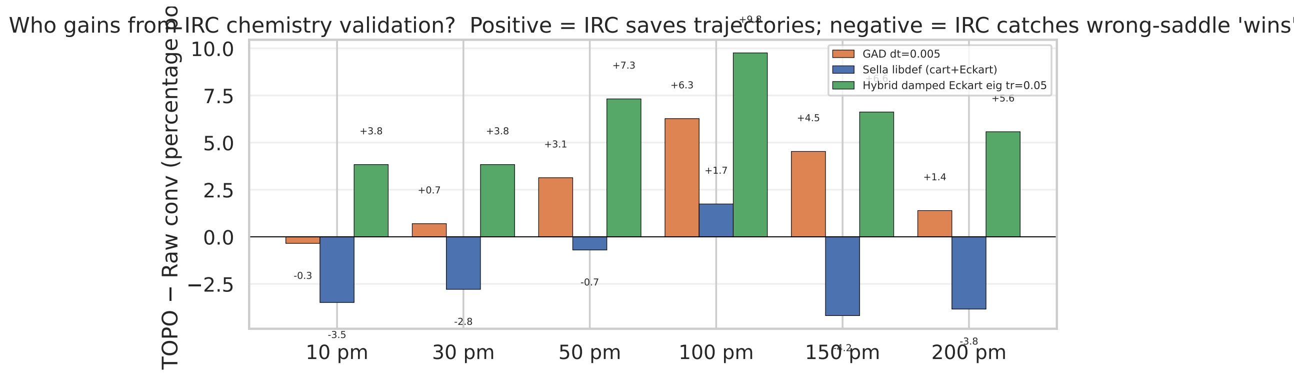

Figure 4: Who gains from IRC chemistry validation? Bars show *IRC TOPO − Raw conv* per (family, noise). Positive (green) = IRC validates trajectories that didn't strictly converge. Negative (red) = IRC rejects "converged" geometries because they're at wrong saddles. **Sella's recovery is negative at 10 and ≥150 pm** — IRC catches its wrong-saddle "wins". Hybrid and plain GAD gain consistently. The hybrid has the largest recovery (+3.8 to +9.8 pp).

Mechanism: basin character vs strict threshold. GAD walks slowly through the right basin; even when raw conv fails (force plateaus at $f_{\max} \approx 0.01$), IRC validates the geometry. Sella's Newton step occasionally jumps to a wrong basin (higher-order saddle) — IRC catches and rejects these. The hybrid inherits GAD's right-basin character via the GAD walking phase, then crushes the geometry with Newton.

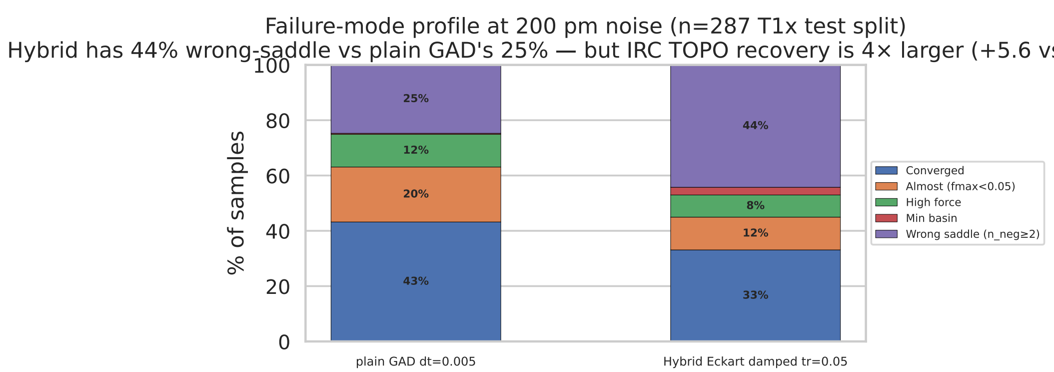

Figure 5: Failure-mode profile at 200 pm. Hybrid pays a wrong-saddle penalty (44% vs plain GAD's 25%) when Newton's eigenvector-following lands on the wrong mode. But many of those wrong-saddle geometries are still in the right reaction basin, so IRC recovers them. Result: hybrid trails plain GAD by 6 pp on IRC TOPO at 200 pm, not 19 pp as raw saddle character suggests.

4 Geometric quality: distance to the labelled T1x TS

The hybrid wins this axis decisively at high noise. Even plain GAD's median converged TS at 200 pm sits 0.01 Å off; Sella's p95 stretches out to 0.8 Å (i.e. some "converged" Sella TSs are an Ångström away from the labelled TS).

Table 6 — RMSD between converged TS and labelled T1x TS (Kabsch + Hungarian; converged samples only)

noise	Sella libdef med (Å)	p95 (Å)	Plain GAD dt=0.005 med (Å)	p95 (Å)	Hybrid damped Eckart eig tr=0.05 med (Å)	p95 (Å)
10 pm	0.008	0.073	0.005	0.018	0.007	0.047
30 pm	0.009	0.071	0.008	0.021	0.007	0.047
50 pm	0.009	0.072	0.011	0.028	0.007	0.049
100 pm	0.009	0.201	0.014	0.044	**0.008**	**0.055**
150 pm	0.013	0.617	0.016	0.088	**0.007**	**0.062**
200 pm	0.017	**0.838**	0.014	**0.456**	**0.008**	**0.109**

At 200 pm: hybrid is 7.7× tighter p95 than Sella and 4.2× tighter than plain GAD. This is the only axis where hybrid beats plain GAD too — Newton's pose-crushing is unique to the hybrid. Source: `analysis_2026_04_29/rmsd_to_known_ts_compare.csv`.

5 Hybrid mechanism: three Newton-firing regimes

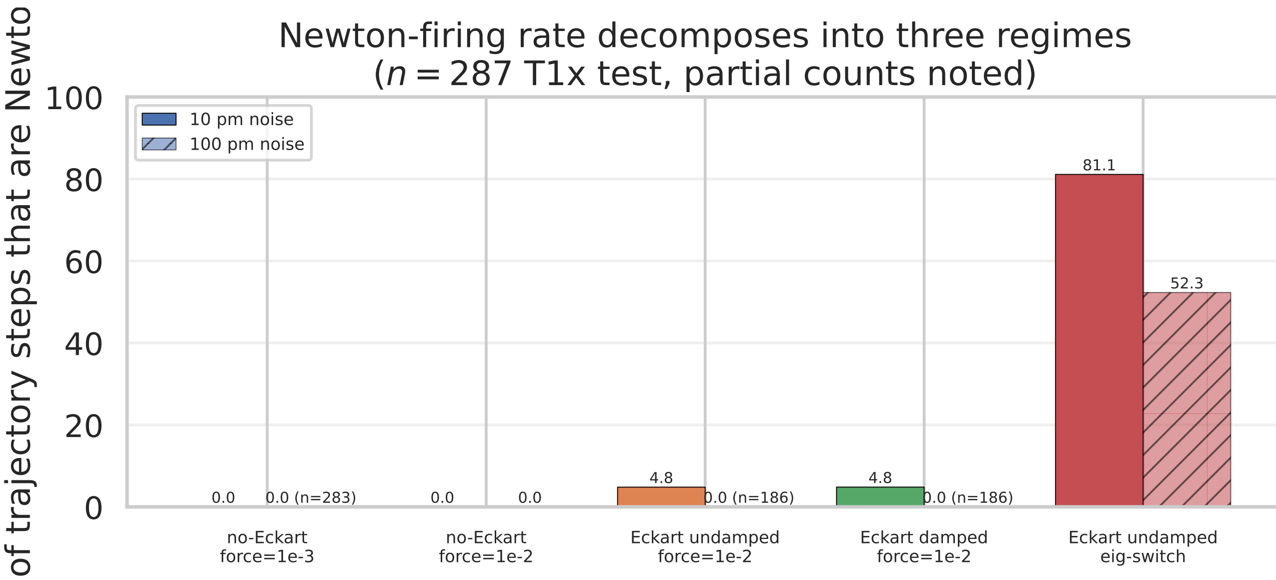

Figure 6: Three regimes of Newton activity. Per-step Newton fraction by config × noise. (1) Never: no-Eckart with sf ≤0.01, Cartesian force-norm never crosses threshold. (2) Sometimes: Eckart force-switch sf=0.01 fires Newton 5% of steps at 10 pm, 0% at 100 pm. (3) Always: Eckart eigenvalue-switch fires Newton 52–81% of steps at all noise levels. *Only eigenvalue-switch reaches Newton reliably at high noise.*

Mechanism summary.

- **No-Eckart hybrid never fires Newton at the project-default switch threshold (10^{-3}).** The Cartesian force-norm $\|F\|_2$ at the GAD plateau ($f_{\max} \approx 0.01$) sits at 0.03–0.1 for a 6-atom T1x molecule — far above 10^{-3}. Confirmed by placebo test: sf=10^{-3} vs sf=10^{-2} give identical raw conv at 100 pm (67.6%, 67.6%) because Newton never fires under either.
- **Eckart force-switch fires Newton at low noise only.** Internal-coord force is smaller (TR modes removed); reaches sf=0.01 on 5% of trajectory steps at 10 pm but essentially never at 100 pm.
- **Eckart eigenvalue-switch fires Newton reliably at every noise level** (52–81% of steps). Newton fires when the Hessian's vibrational eigenvalues match a Morse-index-1 signature, regardless of force magnitude. This is the only configuration that engages the Newton phase across the full noise range.

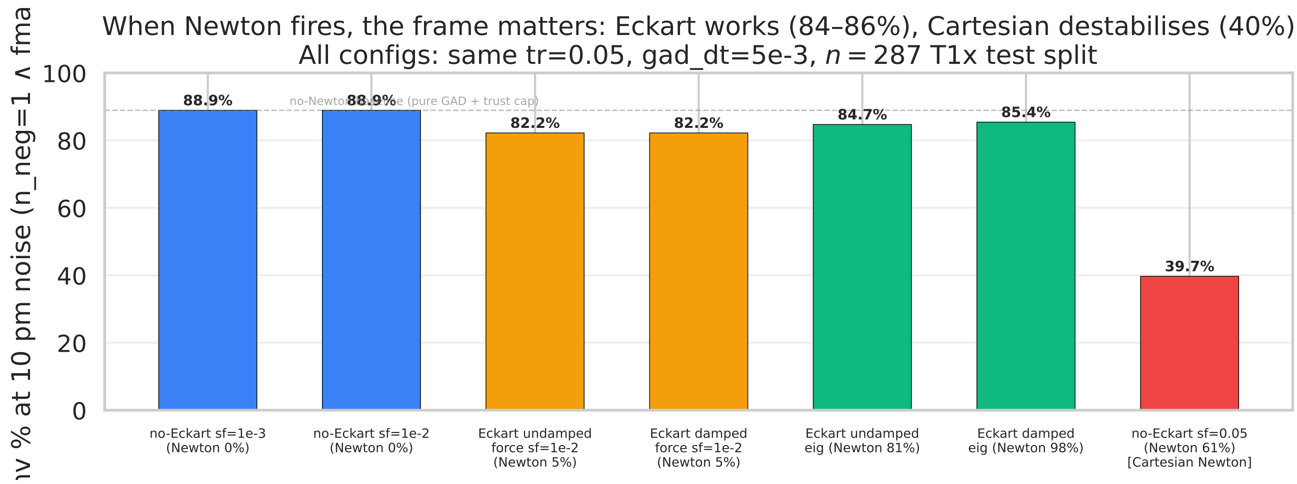

Figure 7: Frame matters when Newton fires. 7 configurations at 10 pm, color-coded by Newton-firing regime. Red bar = no-Eckart with sf=0.05 (Cartesian Newton fires 61% of steps) — raw conv crashes to 39.7%, but **IRC TOPO is still 88.9%**, tied with all other configs. Cartesian Newton displaces the geometry by 0.3–1 Å (RMSD- intended drops to 26% vs other configs' 45–46%) but keeps the trajectory in the right basin. *Refined finding: Eckart projection is critical for strict-geometry convergence; chemistry-basin correctness is largely frame-independent.*

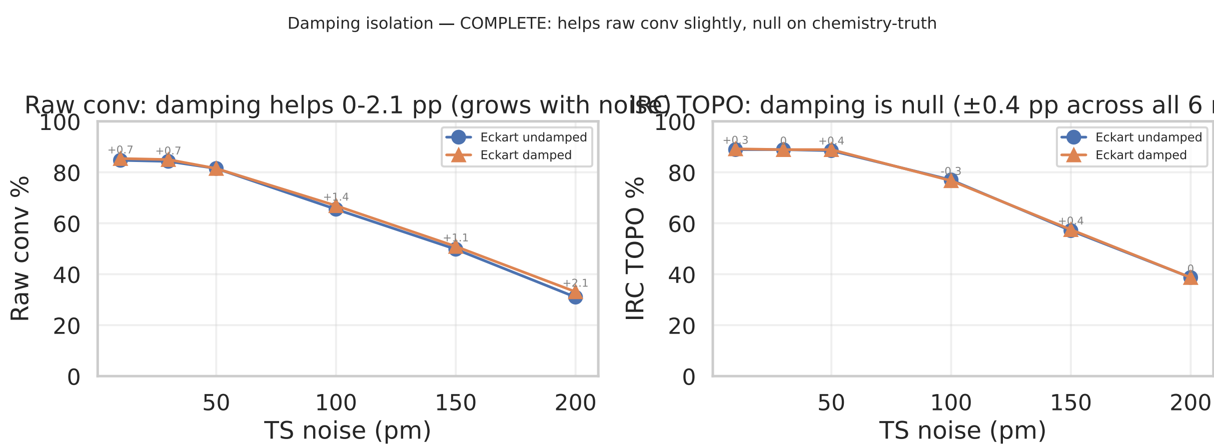

Figure 8: Damping helps raw conv slightly, null on chemistry. *Left:* damped Eckart eig-switch beats undamped by 0–2.1 pp on raw conv at every noise, with the benefit growing monotonically with noise. *Right:* damping is ±0.4 pp on IRC TOPO at every noise — null. Mechanism: damping ($|\lambda_i|$ floor at `min_curvature`) prevents tiny eigenvalues from generating divergent Newton steps, which matters more at high noise where eigenvalues fluctuate. But it doesn't change which basin the trajectory ends up in. Use damped — never hurts, helps a hair on raw conv.

6 Convergence-criterion mismatch: raw conv ≠ chemistry

The 6 hybrid configurations at 100 pm range from 1.7% to 66.9% raw conv (a 65 pp spread). On IRC TOPO they range from 73.2% to 79.8% (a 7 pp spread). **The strict $f_{\max} < 0.01$ threshold dramatically overstates the chemistry-relevant difference.**

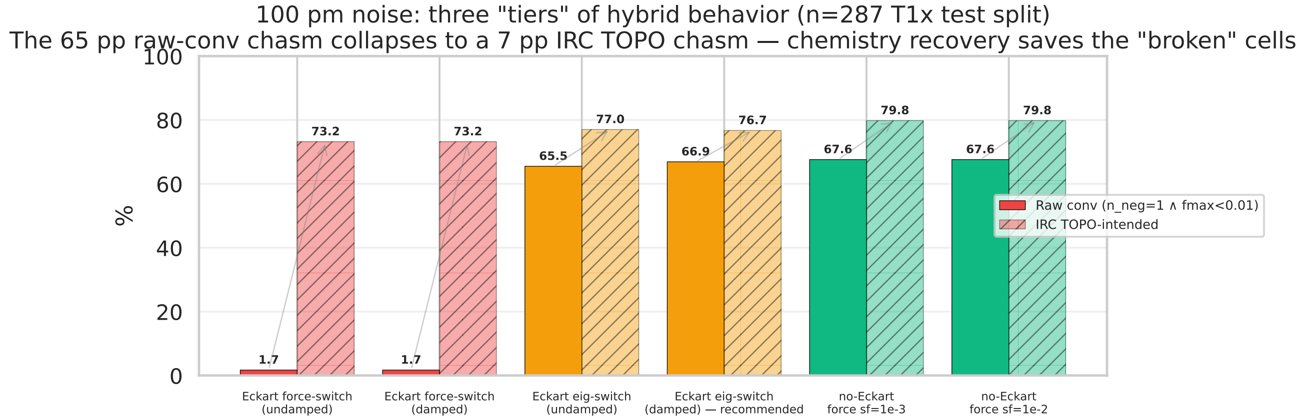

Figure 9: The 65 pp raw-conv chasm collapses to a 7 pp IRC TOPO chasm at 100 pm. Three tiers: (red) Eckart force-switch (Newton barely fires, 1.7% raw, +71.5 pp recovery); (orange) Eckart eig-switch (Newton fires often, mid recovery); (green) no-Eckart force-switch (GAD-with-trust-cap, high raw conv and TOPO). *Convergence threshold strictness ≠ chemistry basin correctness.* A trajectory can stay in the right basin for 1000 steps without ever crossing $f_{\max} < 0.01$.

Open questions / future work

Documented in `BACKBURNER.md` §Z1–Z4:

- **Z1.** Mode-overlap-aware switching: gate Newton trigger on eigenvector continuity in addition to n_{neg}=1. Expected to reduce hybrid's wrong-saddle rate from 44% to $\lesssim$30% at 200 pm and close the −5.9 pp IRC TOPO gap vs plain GAD.
- **Z2.** Hybrid noise sweep at 300/500/1000 pm: extrapolate high-noise trend. Sella's n_{conv} collapses there; the hybrid's wall advantage should widen.
- **Z3.** Full Cartesian Newton sf sweep {0.03, 0.05, 0.10} on no-Eckart: characterize the "raw conv crashes / TOPO preserved" trade-off across thresholds. Curiosity item.
- **Z4.** Rigorous IRC integrator (vs Sella IRC) for chemistry validation: tests whether the chemistry-recovery effect is IRC-algorithm-dependent.

Sources & reproducibility

- **Master data table:** `analysis_2026_04_29/master_2026_05_11.csv`
- **Comprehensive findings catalog:** `HYBRID_FINDINGS_CATALOG.md` (1180+ lines)
- **Per-cell summaries:** `runs/hybrid_for_irc/`, `runs/hybrid_deeper/`, `runs/hybrid_extension/`, `runs/test_dtgrid/`, `runs/test_hessfreq/`, `test_summary_full.csv`
- **IRC validation parquets:** `runs/irc_hybrid/`, `runs/irc_hybrid_deeper/`, `runs/irc_hybrid_extension/`, `runs/test_irc/`
- **Hybrid step functions (canonical):** `src/gadplus/search/hybrid_gad_damped_eigfollownewton_eckart.py` `src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py`, `src/gadplus/search/hybrid_gad_eigfollow`
- **Standalone runner:** `scripts/hybrid_gad_newton_runner.py`
- **Figure-build script:** `scripts/build_pdf_2026_05_11.py`

- **Trajectory diagnostics (catalog appendix):** step-size distributions, time-to-first-Newton, Morse-1 retention, TOPO vs RMSD disagreement, IRC endpoint stability, force-descent "Newton cliff", wrong-saddle scaling.